EUNOIA JUNIOR COLLEGE
JC2 Preliminary Examination 2022
General Certificate of Education Advanced Level
Higher 2

CANDIDATE NAME	**ANSWER KEY**
CIVICS GROUP	2 1 -
REGISTRATION NUMBER	

H2 Biology 9744/03

Paper 3 Structured & Free Response Questions

19 September 2022

2 hours

Candidates answer **Section A** on the Question Paper and **Section B** on the Answer Booklet.

Additional Materials: 12-page Answer Booklet

READ THESE INSTRUCTIONS FIRST

Write your name, civics group and registration number on all the work you hand in.
Write in dark blue or black pen on both sides of the paper.
You may use an HB pencil for any diagrams or graphs.
Do not use paper clips, highlighters, glue or correction fluid/tape.

Section A
Answer **all** questions.

Section B
Answer **one** question on the **12-page Answer Booklet**.
Write your answer to each part of the question on a fresh sheet of paper.

The use of an approved scientific calculator is expected, where appropriate.

The number of marks is given in brackets [ ] at the end of each question or part question.

At the end of the examination, ensure that you submit both the question paper and answer booklets.

For Examiner's Use	
Section A	
1	
2	
3	
Section C	
4 OR 5	
Total	**75**

This document consists of 24 printed pages and **1** blank page.

Section A

1 (a) Influenza causes occasional pandemics that have claimed the lives of millions of people.

Different strains of influenza are named in terms of 'H' and 'N'. For example, avian influenza is named H5N1.

Fig. 1.1 shows the main structural features of the H5N1 influenza virus.

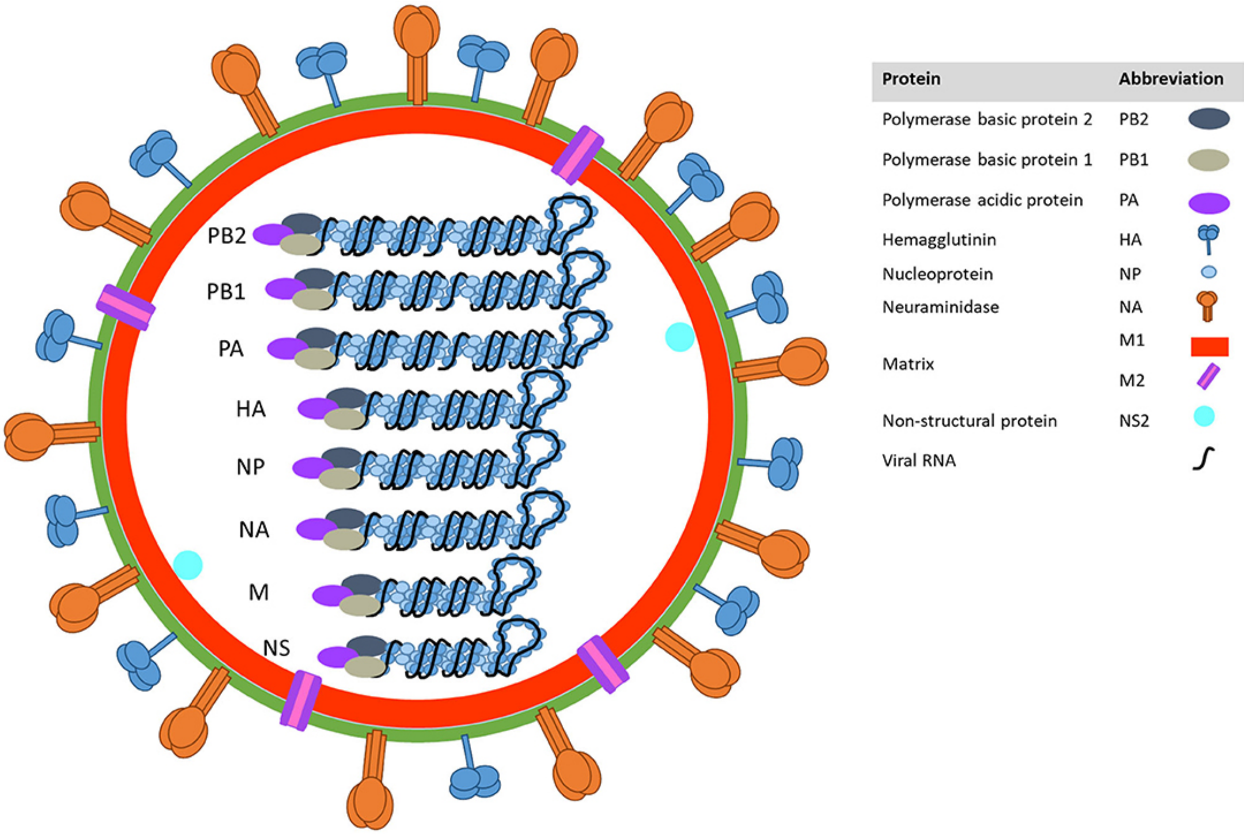

Fig. 1.1

(i) Explain the functions of haemagglutinin and neuraminidase in an enveloped virus, such as H5N1.

Haemagglutinin

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.... [1]

1. **Haemagglutinin** **binds** to **sialic acid** receptor on host cell membrane, facilitates entry into host cell by **endocytosis**.

Neuraminidase

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.... [1]

2. **Neuraminidase** **cleaves** the **sialic acid** to facilitate **release** / **budding** of the viruses from the infected cells.

(ii) Other than components stated in Fig. 1.1, name two other components of the envelope of a virus, such as H5N1.

.... [1]

1. **Phospholipids** and **cholesterol.**

(iii) The emergence of new strains of influenza continues to pose challenges to public health and the scientific communities.

Fig. 1.2 shows various ways that new strains of influenza, such as H5N1, may arise.

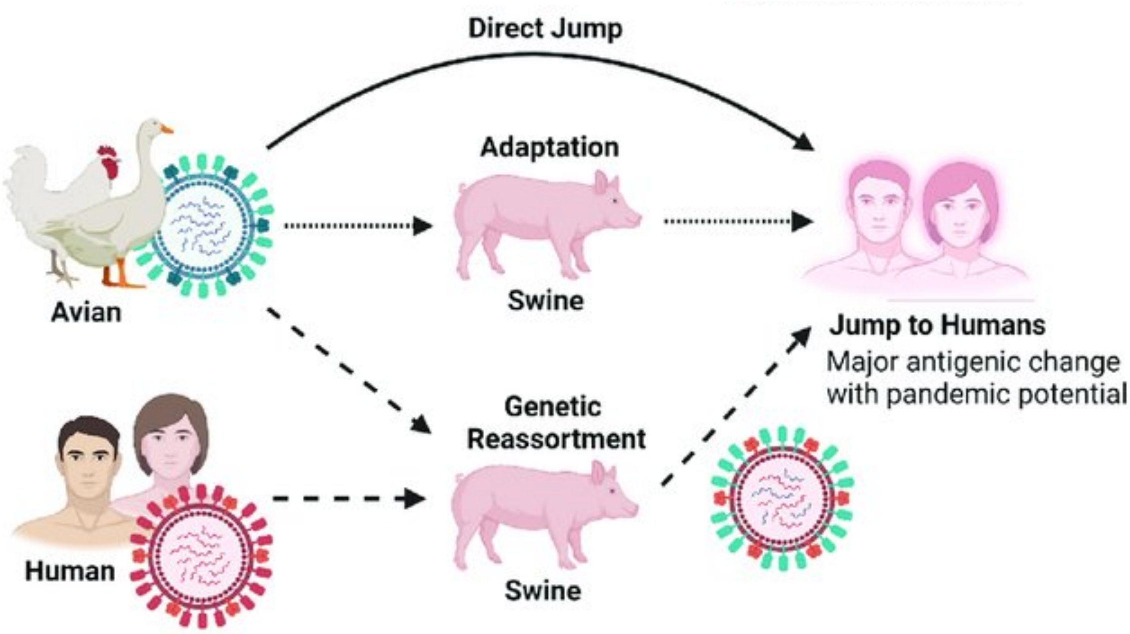

Fig. 1.2

Outline how a new strain may arise due to genetic reassortment.

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.... [3]

1. New viral strains are also formed when **two or more strains** of influenza viruses from **avian and human infect a swine host** where
2. **reassortment** of the different **RNA segments** occur resulting in a **new combination of RNA segments** in a **virion**, giving rise to **antigenic shift**.
3. **New combination of hemagglutinin and neuraminidases** at the viral envelope.

(iv) Suggest how adaptation of influenza virus in swine, as seen in Fig. 1.2, may have occur.

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…………………………………………………………………………………………………… [2]

1. There are **accumulation of mutations** of the **viral genome** leading to the changes in the ribonucleotide sequence (I: antigenic shift)
2. as a result of the **lack of proofreading** ability of **RNA-dependent RNA polymerase** and the **fast/high rate of replication** of the virus.

(v) An individual's immune responses can change throughout their lifetime.

Fig. 1.3 shows one person's immune response to the influenza virus when they were first infected and when they were infected two years later by a new, mutated strain of the virus.

The influenza virus has many antigens to which the immune system can respond. Fig. 1.3 shows the response to four of these antigens (A–D).

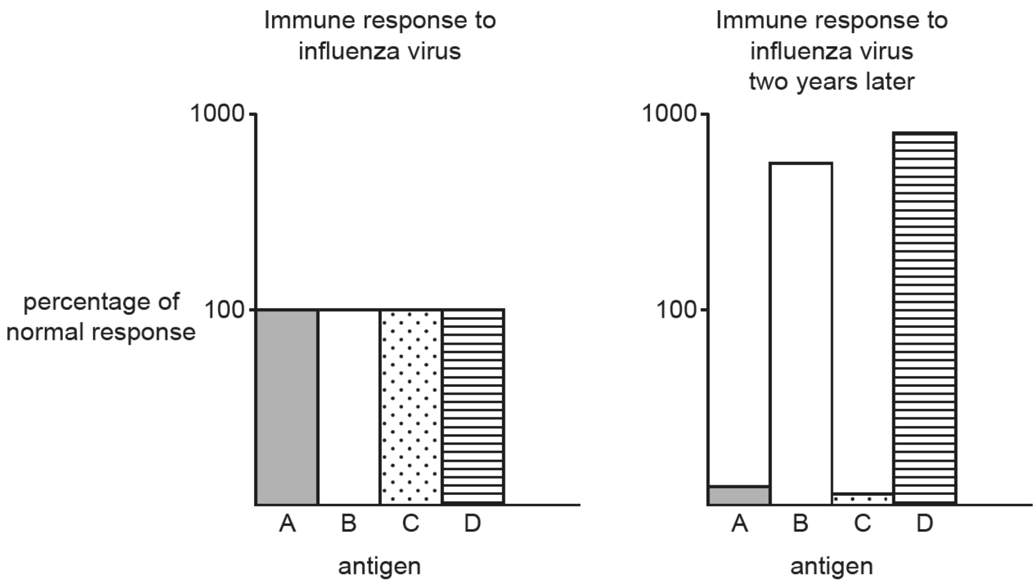

Fig. 1.3

Explain the differences in the person's initial immune response to the influenza virus with their immune response two years later.

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1. **Memory cells** produce a **stronger / larger response** to **antigens B and D**. NOTE: context is generally immune response, no need to specify memory B or T cells.
2. **Mutated virus** has **less of / no longer has antigens A and C**.

(b) Tuberculosis (TB) is another respiratory infectious disease that affects human.

The vaccine used to control TB is known as Bacillus Calmette-Guérin (BCG). The vaccine contains live bacteria that have been selected so that they do not cause disease in humans.

Fig. 1.4 shows a macrophage that is in the process of engulfing the bacteria in the vaccine.

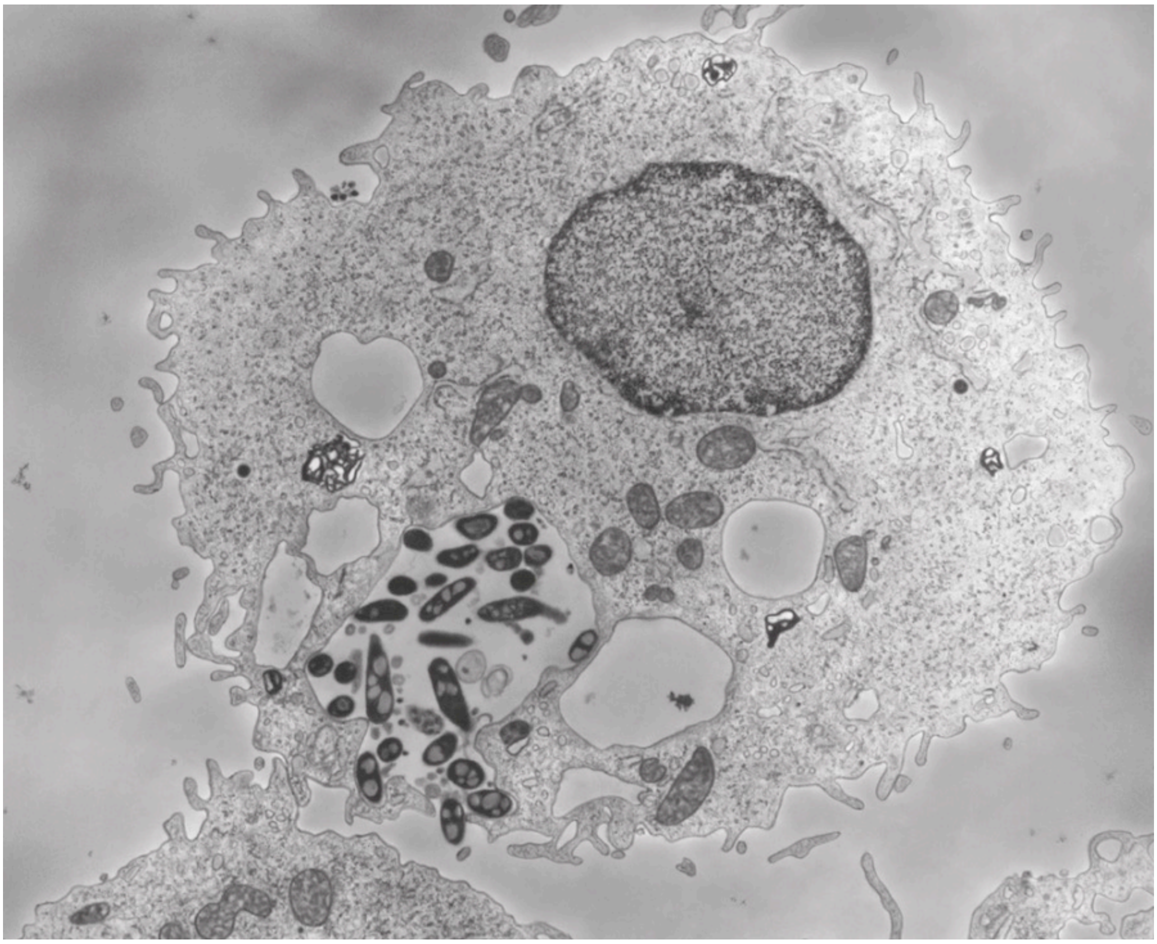

Fig. 1.4

(i) Name the pathogen that causes TB.

.... [1]

1. ***Mycobacterium tuberculosis*** (A: without underline)

(ii) Describe how this pathogen is transmitted.

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.... [2]

1. When an **infected** person with the active TB disease **sneezes or coughs**,
2. and an **uninfected** person **inhales** the **fine, droplets** containing the bacteria.

(iii) Outline the events that occur in the body after the macrophage has engulfed the bacteria until the production of antibodies in response to the BCG vaccine.

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.... [5]

1. **Fusion** of phagocytic **vesicle with lysosomes**, **hydrolysis of bacteria** by hydrolytic enzymes
2. **Processing antigens** to form **antigen-MHC complex** to be presented on cell surface membrane to become **antigen presenting cell** (APC).
3. Binding of APC to **naïve T cell** results in **secretion of cytokines** that will **activate** the naïve **T cells** which will undergo clonal expansion and differentiation to form effector and memory T cells. (A: T helper cells)
4. **Naïve B cells** encounters vaccine/bacteria and also **present the antigen** on its cell surface membrane, which is **recognised** and bound by antigen-specific **T helper cell**.
5. **T helper cells** secrete cytokines that **stimulate / activate** specific **naïve B cells** to become **antibody-secreting plasma cells**.

(iv) The treatment for people with active TB lasts six months and involves a combination of antibiotics. This is usually very effective if the person has a susceptible (non-resistant) strain of the bacteria.

Table 1.1 summarises one recommended treatment strategy that involves a combination of antibiotics.

Table 1.1

antibiotic	length of treatment	mode of action of antibiotic
rifampicin (R)	6 months	enters bacterial cells and inhibits protein synthesis
isoniazid (H)	6 months	prevents the synthesis of cell wall components known as mycolic acids
ethambutol (E)	first two months	prevents mycolic acids from being added to the cell wall
pyrazinamide (Z)	first two months	prevents the synthesis of fatty acids

Susceptible strains of the bacteria will be killed using any one of the antibiotics listed in Table 1.1. However, combination treatment is preferred as it is one method that can be used to reduce the impact to society of antibiotic resistance.

With reference to Table 1.1, explain how combination treatment for TB can help to reduce the impact of antibiotic resistance compared to single antibiotic treatment.

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Combination treatment:

1. **acts on different targets** / have **different modes of actions** to **kill all** the bacteria.
2. **resistance** / **mutation unlikely to occur on all antibiotics** at the same time, hence there will be some antibiotics that are **still effective** (OWTTE).
3. **longer treatment time increases chance of killing all** bacteria
4. **likely to eliminate bacteria more quickly** so less resistance can occur
5. AVP

[Any two]

(v) Rifampicin binds tightly to an RNA polymerase molecule close to its active site. This affects the activity of the enzyme.

During the formation of RNA, a number of events occur that involve the action of RNA polymerase.

Suggest two ways in which rifampicin can affect the activity of RNA polymerase.

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1. It **alters the conformation** of the **active site**, resulting in **substrate / nucleotides not able to bind** to active site. (I: non-competitive inhibitor / allosteric inhibitor) (R: competitive inhibitor) NOTE: close to, not at active site.
2. **Complementary base pairing between DNA and RNA cannot / less easily formed**.
3. It **prevents / reduces frequency of attachment to promoter**, and **no / less likely to initiate transcription**. (A: description of transcription initiation)
4. It **prevents / reduces frequency of mRNA elongation** due to **no / less likely for phosphodiester bonds to form.**
5. **AVP**

[Any two]

(vi) Countries are classified by the World Bank into one of four income groups.

Table 1.2 shows the estimated incidence of TB for 2012 to 2016 for these income groups.

The incidence represents the number of new cases of TB occurring per 100 000 people in one year. The new cases include the number of cases that have occurred again after a period of recovery (relapse TB).

Table 1.2

	incidence per 100 000 people				
year / **income group**	**2012**	**2013**	**2014**	**2015**	**2016**
low	253	244	238	231	224
lower middle	244	240	236	232	227
upper middle	84	81	78	76	74
high	14	13	13	12	12

Describe the patterns and trends shown in Table 1.2.

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.... [2]

1. There is a **decrease in incidence over time** for **all groups** e.g., in low income group from 253 in 2012 to 224 in 2016 (A: any relevant data from 2012 to 2016).
2. **Similar incidence rate** between **low and lower middle income** group e.g., 224 in low and 227 in lower middle in 2016 (A: any relevant data from 2012 to 2016).
3. There is a **lower incidence rate in upper middle** / **high** income group than **low** / **lower middle** income group (ORA), e.g., 74 in upper middle and 227 in lower middle in 2016 (A: any relevant data from 2012 to 2016).
4. There is a general decrease in incidence rate with increase in income group e.g., 253 in low to 14 in high in 2012 (A: any relevant data from 2012 to 2016).
5. AVP

[Any two]

(vii) In general, the countries that do not have a BCG vaccination programme are high-income countries that have a low number of cases of TB. In most of these countries, the vaccine is given only to babies and children at high risk of developing TB.

Suggest **one** reason why a child in a country with a low number of cases of the disease could be at a high risk of developing TB.

....

.... [1]

1. Children with **compromised immune system** which makes them more susceptible to infection (OWTTE).
2. They have family members that are from a country that has high TB incidences (OWTTE)
3. They are travelling to / recently returned from countries that have high incidences of TB.
4. They live in an area that has outbreak of TB.
5. AVP

[Any one]

NOTE: the question is asking why babies who live in low incidence rate country could be at **high risk** of developing TB. It's not referring to general population, hence herd immunity is not relevant in this question.

(c) Sucrose phosphorylase is an enzyme found in some species of bacteria. One function of this enzyme is for the production of compounds that help to protect the cell from harmful osmotic changes in the external environment.

(i) Fig. 1.5 shows the reversible reaction that takes place within the bacterial cell.

$$\text{sucrose} + \underset{\text{inorganic phosphate}}{P_i} \xrightleftharpoons{\text{sucrose phosphorylase}} \alpha\text{-glucose-1-phosphate} + \underset{\text{reducing sugar}}{\mathbf{X}}$$

Fig. 1.5

Name reducing sugar **X** in Fig. 1.5.

.... [1]

1. **Fructose**

(ii) In the absence of sucrose phosphorylase as a catalyst, the reaction shown in Fig. 1.5 would take too long to occur to allow the bacterial cell to function efficiently.

Explain why the reaction shown in Fig. 1.5 proceeds at a much faster rate in the presence of the enzyme.

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.... [1]

1. Enzyme **<u>lowers the activation energy</u>** of the reaction by… (mechanism)…

(iii) An enzyme that catalyses a reaction of commercial interest needs to be investigated to see if it is suitable for use in industry.
For example:

- immobilised enzymes may be used as they have a longer shelf-life than the enzyme free in solution
- many industrial reactions are carried out at higher temperatures to minimize contamination of products by microorganisms.

Fig. 1.6 shows the results of an investigation to compare the activity of sucrose phosphorylase free in solution (free enzyme) with immobilised sucrose phosphorylase (immobilised enzyme) at different pHs.

Fig. 1.7 shows the activity of the free enzyme and immobilised enzyme at different temperatures.

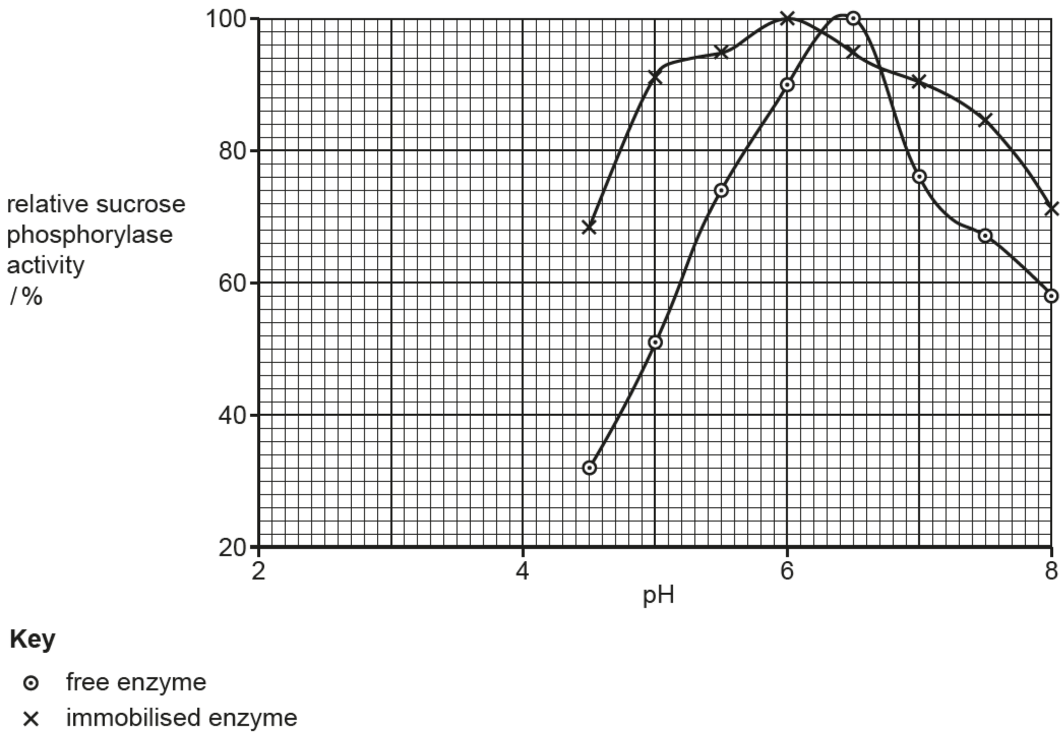

Fig. 1.6

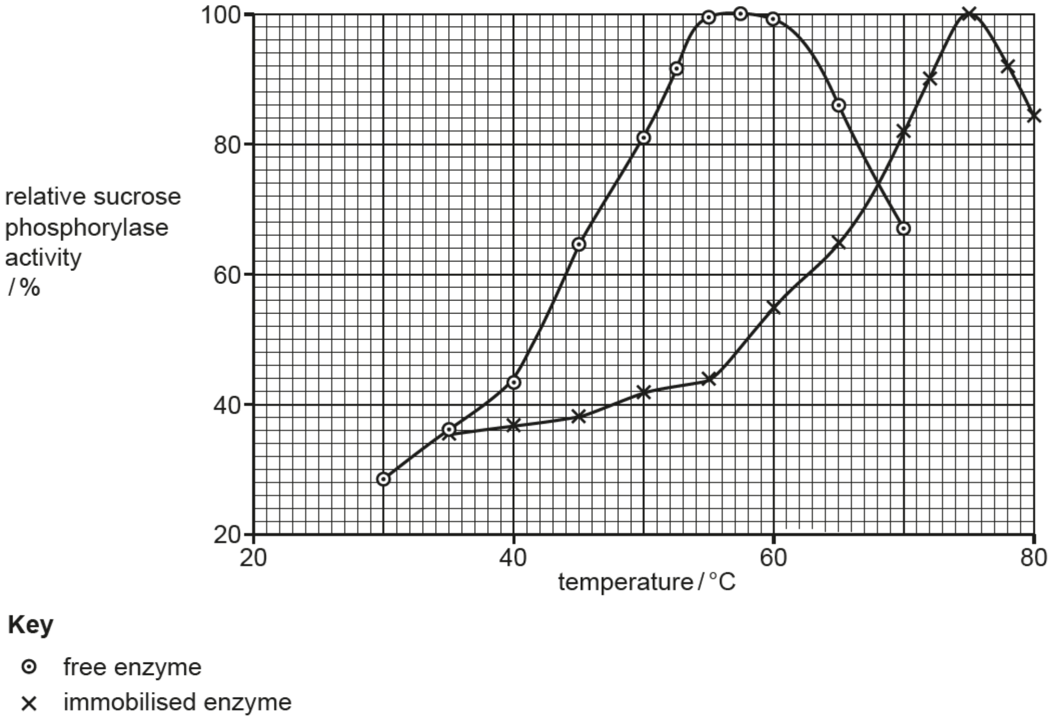

Fig. 1.7

With reference to the results shown in Fig. 1.6 and Fig. 1.7, discuss why **immobilised** sucrose phosphorylase enzyme could be better for use in industrial reactions.

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.... [3]

1. It has a **higher optimum temperature** hence more **suitable for reactions that require higher temperatures / thermostable / have a longer shelf-life (enzymes can be reused more often) / less microbial contamination due to high heat** (OWTTE).
2. In the same pH range, it has a **greater range of pH with higher activity / greater stability** hence **less affected by pH changes** on its enzyme activity (OWTTE) (A: resists changes in pH)
3. Data citation: 1 set of relevant data for temperature and pH (e.g. **75 °C vs 56 – 58 °C / 17°C higher** and **68% activity or higher between pH 5 to 8**)

[Total: 30]

2 Researchers investigated the extent to which the founder effect and natural selection affected evolutionary change.

Fig. 2.1 shows the brown anole lizard, *Anolis sagrei*. These lizards live on a number of Caribbean islands and feed on a variety of invertebrates and other small animals.

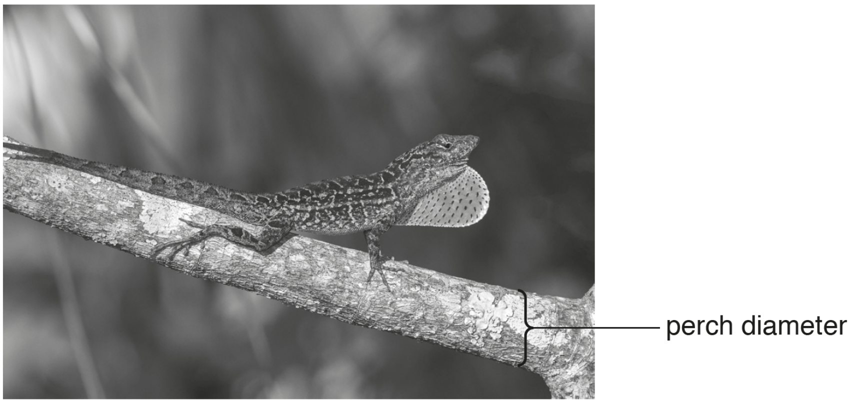

Fig. 2.1

A. sagrei spends a lot of time perching (resting) on, or moving along, branches of shrubs and trees. The width of the branch that *A. sagrei* perches on is known as the perch diameter, as labelled in Fig. 2.1.

There is a positive correlation between perch diameter and hind limb length of *A. sagrei*.

- Longer hind limbs allow *A. sagrei* to run faster on vegetation with a larger diameter.
- Shorter hind limbs are needed to provide stability on vegetation of a smaller diameter.

In 2004, a hurricane caused the death of all the *A. sagrei* lizards on seven islands.

In 2005, the researchers randomly collected seven male and seven female lizards from a source population on a nearby island. For each of the seven islands affected by the hurricane, a male and female lizard were mated and placed on each island. These islands formed the experimental founder islands where new populations of *A. sagrei* were successfully established from each founding pair.

Fig. 2.2 shows the difference in vegetation between the source island and the seven experimental founder islands.

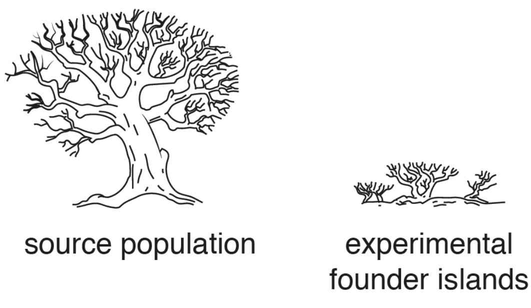

Fig. 2.2

(a) **(i)** Predict the effect of natural selection on mean hind limb length of *A. sagrei* on the seven experimental founder islands.

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1. (mean hind limb length) should **decrease** / legs get **shorter**.

(ii) Predict how collecting individuals at random for the seven founding pairs affects the mean hind limb length of *A. sagrei* on the different islands.

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1. (mean hind limb length) will **vary;**

(b) Many generations of *A. sagrei* were produced over the four years after the introduction of the founding pairs.

Fig. 2.3 shows how the mean hind limb length of *A. sagrei* changed on the seven experimental islands and on the source island.

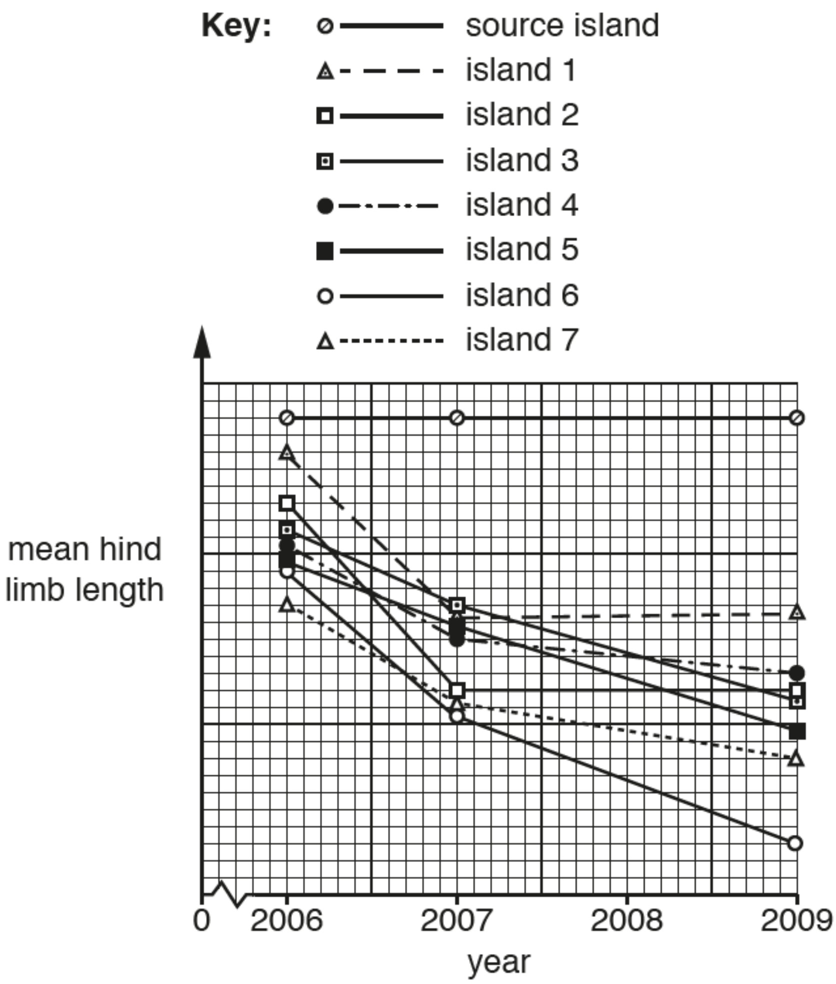

Fig. 2.3

With reference to Fig. 2.2 and Fig. 2.3, describe **and** suggest explanations for the results for the islands.

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.... [5]

1. There is **directional selection** as the **hind limb length decreased** in **all** / **seven** **experimental** / **founder** **islands**. (R: all islands, because data includes source island, which is *does not* have decreased hind limb length)
2. There is **stabilising selection** as the **hind limb length stayed the same** on **source island**.
3. [Selection pressure] The **selection pressure** is **reduced** / **thinner perch diameter** (A: branch / perch).
4. [Selective Advantage & Example] Individuals with favourable traits such as **shorter limbs** have a **selective advantage** and are **selected for** (R: favourable alleles are selected for') because there is **increased stability**.
5. [Differential survival & reproduction] Individuals with shorter limbs were selected for and so **survived and reproduced to produce fertile, viable offspring**; **Favorable alleles were passed to offspring** resulting in **higher frequencies of the alleles in the descendent population over time**. (R: 'passing favourable traits')

NOTE: point on variation not important in this context since it is founder effect.

(c) In the investigation, one population of *A. sagrei* was established on each experimental founder island.

Outline how speciation may occur on the seven experimental founder islands.

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.... [3]

1. There is **allopatric speciation.**
2. The **islands** are **geographically isolated** as they are surrounded by water that acts as a physical barrier preventing interbreeding. This results in the **disruption of gene flow;**
3. Over **many generations,** each population of *A. sagrei* **evolve independently on different islands** (with change in allele frequencies due to natural selection, genetic drift and accumulation of genetic mutations), became **reproductively isolated** and can no longer interbreed to produce viable, fertile offspring**.**

Accept other points:

4. **Different mutations** may occur to the populations in the seven islands, resulting in **different variation**.
5. Different islands will present **different selection pressures** and individuals best adapted to the environment (or individuals with favourable trait) will have a **selective advantage** will be **selected for** (they survive to reproduce) and **favourable alleles will be passed on to the next generation** and so the **frequency of favourable alleles will increase**.

[Total: 10]

3 **(a)** Fig. 3.1 shows the structure of a prokaryotic cell.

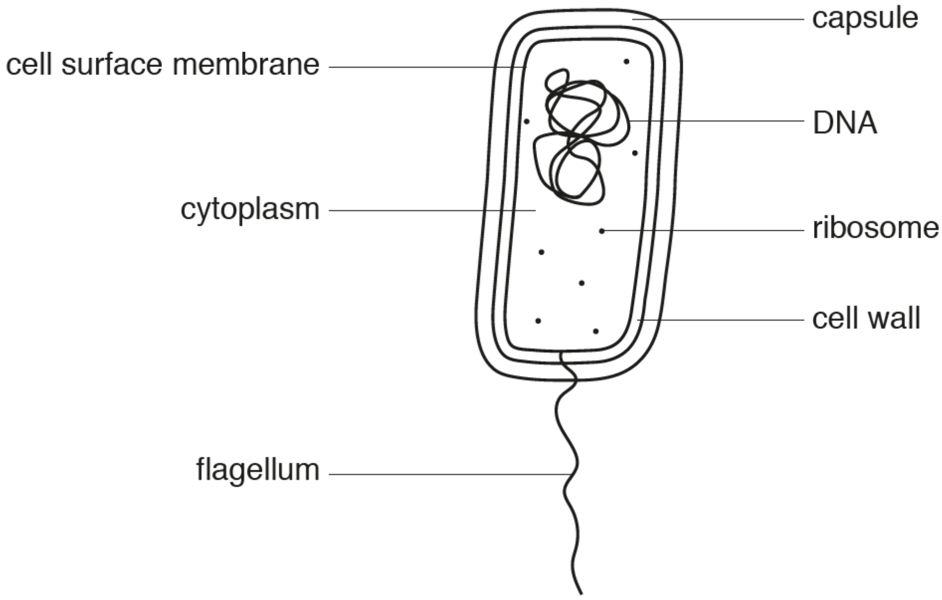

Fig. 3.1

Fig. 3.1 has not been fully labelled to confirm that the cell is prokaryotic.

State what other information could be added to two of the labels to confirm that this cell is prokaryotic and not eukaryotic.

....

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.... [1]

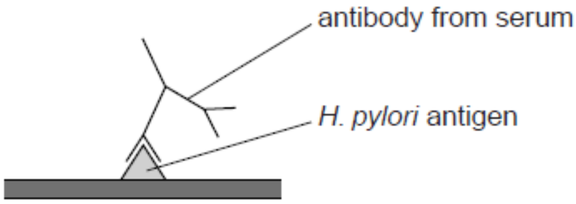
1. **70S** ribosomes
2. **Circular** DNA / **non-histone** DNA / DNA not **enclosed in nucleus**
3. **Peptidoglycan** cell wall
4. AVP

[Any two points]

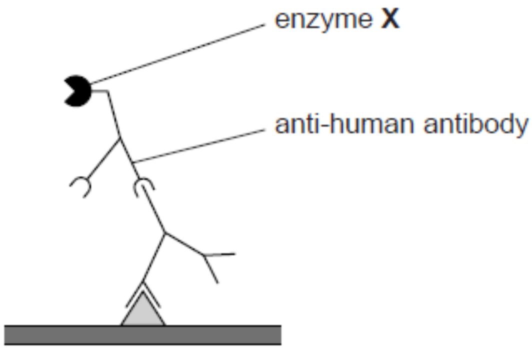
(b) The bacterium *Helicobacter pylori* has been associated with diseases such as gastric ulcers and stomach cancer.

An infection with *H. pylori* can be diagnosed by testing for the antibodies produced in response to *H. pylori* antigens, as shown in Fig. 3.2.

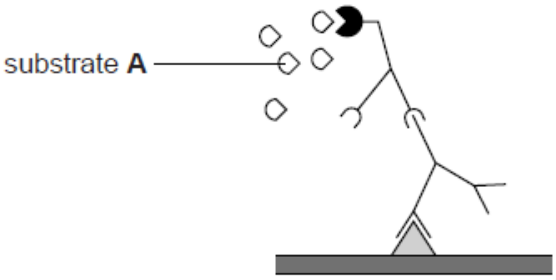
step 1
add blood serum samples to antigens of *H. pylori* attached to wells in a testing plate



step 2
rinse the testing plate and add anti-human (secondary) antibody linked to enzyme **X**



step 3
rinse the testing plate and add substrate **A**, which is converted to a coloured product by enzyme **X**



step 4
reaction stopped and colour noted

Fig. 3.2

With reference to Fig. 3.2, explain the importance of the following in this test:

(i) the *H. pylori* antigen.

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1. the **antigen** is **specific** to the **antibody from the serum** and can be used to **detect** the presence of ***H. pylori*** in the patient.

(ii) the use of enzyme **X**

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1. it is able to **catalyse a coloured product** to help **visualize** the presence of antibodies produced in response to *H. pylori*.

(c) Amoxicillin is an antibiotic that is commonly used to treat *H. pylori*, it is similar to penicillin.

Suggest the mode of action of amoxicillin.

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1. it is a **competitive inhibitor** to **enzymes** involved in **peptidoglycan synthesis** (I: transacetylase and A: transacetylase as substitute for idea of enzyme).
2. As a result, there is **inhibition to the formation of cross-links between adjacent chains** of peptidoglycan.
3. Bacterial **cell wall becomes weakened** when bacterial **cells** carry out **division**.
4. When bacteria **take in water by osmosis**, the **increased turgor pressure** against the weakened cell wall causes the bacteria to swell and **lyse**.

[Any two]

(d) In some people, *H. pylori* infection can be difficult to treat because, over time, new strains of bacteria arise that have resistance to commonly used antibiotics. The resistance to antibiotics has been found to spread without the need of a vector or bacterium.

Explain how resistance to antibiotics is spread in a population of *H. pylori*.

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.... [3]

1. It is via the process of bacterial **transformation**.
2. **Fragments** of **DNA** containing **antibiotic resistance gene** (from lysed bacterial cells) in the surrounding medium are taken up by a **competent bacterial cell** via **surface proteins**.
3. The DNA containing the antibiotic resistance gene is **incorporated into the bacterial chromosome/DNA** via **homologous recombination**.

(e) *H. pylori* has an operon that is involved in synthesis of an amino acid, **X**. Fig 3.3 shows the operon and how it is being regulated.

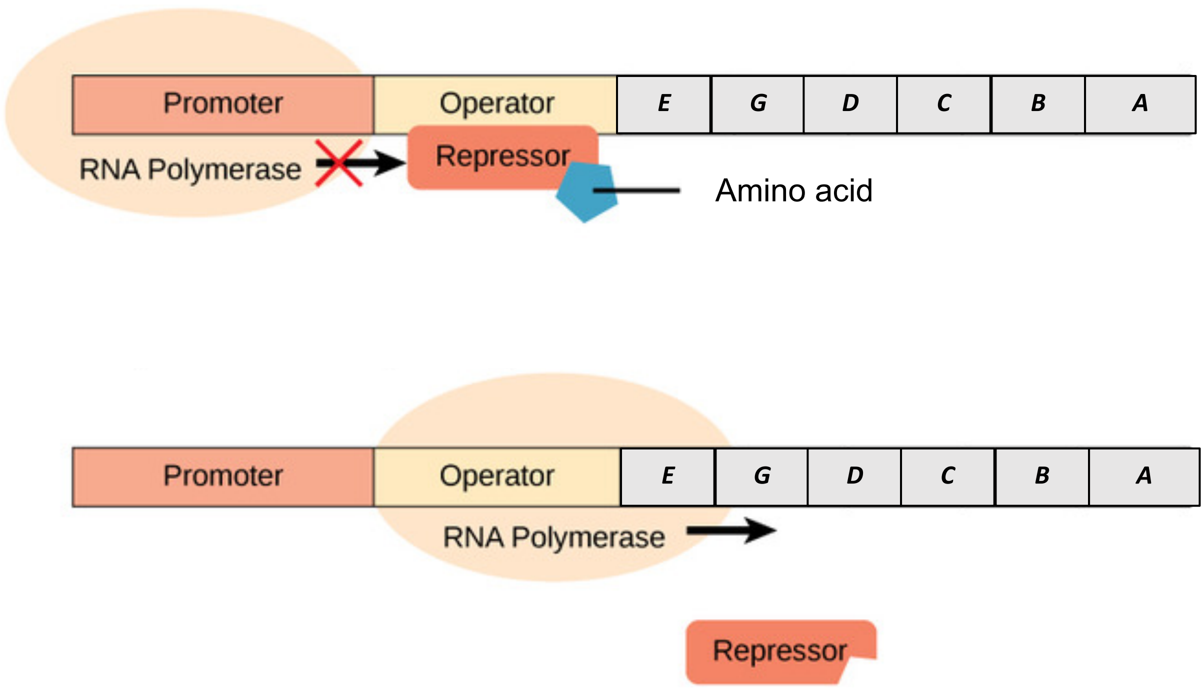

Fig. 3.3

(i) identify the type of operon.

.... [1]

1. **Repressible** operon

(ii) predict the effect of an insertional mutation in the regulatory gene that codes for the repressor.

....

.... [1]

1. [Loss-of-function mutation] **Repressor** will be **non-functional** and there will be **constitutive expression** of the operon.
   OR
2. [Gain-of-function mutation] **Repressor** will be **hyperactive**/**constitutively bound** and the operon is **not expressed**.

[Total: 10]

Section B

Answer **one** question in this section.

Write your answers in the **12-page Answer Booklet** provided.

Your answers should be illustrated by large, clearly labelled diagrams, where appropriate.

Your answers must be in continuous prose, where appropriate.

Your answers must be set out in section **(a)** and **(b)**, as indicated in the question.

4 **(a)** Molecular biology techniques are common methods used in molecular biology, biochemistry, and genetics which generally involve manipulation and analysis of DNA.

Outline the principles of techniques used to analyse DNA. [10]

A. Reasons for Analysing DNA (Brief description) [1]
1. Genetics / Forensics / Mutation and disease detection / Evolutionary studies etc;
2. AVP – e.g. identifying and comparing band patterns between individuals with mutated gene sequences and normal individuals;

B. DNA Extraction (Needed for all techniques) [1]*
3. **DNA** extracted from **blood or saliva** of subject/scene of crime/organisms/fossils etc;

C. Polymerase Chain Reaction [max 1]
4. **Sequence gene/sequence** to identify sequence of interest /gene of interest / mutation etc;
5. **Amplification** of DNA via **polymerase chain reaction**;
6. Use of forward and reverse **primers** to flank the target gene/sequence;
7. Production of **many** copies of gene/sequence of interest for analysis, from **initial small amounts**;

D. Restriction Enzymes [1]*
8. The amplified DNA samples are **digested / cut / hydrolysed** using **restriction enzymes** that cleave at **specific restriction sites**, producing **different fragments**;

E. Gel Electrophoresis [max 4]
9. Different DNA fragments of different sizes are loaded onto an agarose **gel** and subjected to an **electric field**;
10. **Negatively charged** DNA will move towards the **positive electrode** / **anode**;
11. **Meshwork of polymer fibres** that makes up agarose gel **impedes movement** of the DNA fragments, affecting the **longer fragments more than shorter ones**;
12. **Shorter DNA fragments migrate faster** than the **larger DNA fragments** towards the positive electrode;
13. **Separate** a mixture of DNA fragments of different lengths;
14. To visualise the bands, gel slab is next stained with a DNA-binding / **DNA-staining dye**, usually **ethidium bromide**;
15. When the gel is placed under **UV light** afterwards, **DNA bands** will be revealed as the dye bound to the DNA **fluoresces**;
16. Electrophoresis helps to estimate **fragment length/size** – by comparing position of the band relative to bands of the DNA ladder and **relative amount** of DNA;

F. Southern Blotting / Nucleic Acid Hybridisation [max 3]
17. DNA fragments are transferred from the agarose gel slab to a **nitrocellulose membrane**;
18. The bands of DNA fragments bind to the membrane in exactly the **same position** as they were in the gel;
19. The target DNA bands on the membrane can then be **selectively visualised** by using **radioactive probes** to detect **specific DNA sequences**;
20. **Single stranded** DNA probes **hybridise** to specific sequences by **complementary base pairing**;
21. **Autoradiography** is performed by placing an **X-ray film** over the membrane;
22. The radioactivity of the bound probes exposes the film to form an image corresponding to the bands that have base-paired to the probe;

G. AVP e.g. Restriction Fragment Length Polymorphism;

*Required for full credit.

(b) Cycles play important roles in both natural and man-made biological processes.

Write an essay about cycles in biology. [15]

A. Cell Cycle [max 5]
1. The cell cycle is the **sequence of events** which occurs **between the formation of a cell and its division into daughter cells**;
2. Three main stages: **interphase, nuclear division, cytokinesis**;
3. During interphase - Cell produces many materials and or organelles required for carrying out all its functions;
4. Cell **replicates its DNA** (during **S phase of interphase**) to prepare for nuclear division;
5. Nuclear division – **mitosis** – forming **identical** cells for **growth / repair / asexual reproduction** AND **meiosis** – forming **gametes**;
6. Cytokinesis - Division of **cytoplasmic contents** into **2 daughter cells**;
7. **Fusion** of a **haploid** sperm and **haploid** egg during **fertilisation** results in the formation of a **diploid zygote**;
8. After fertilisation, the zygote undergoes a process of nuclear division **(mitosis)**; **(**This generates cells that are **genetically identical** to the original zygote)

B. Krebs Cycle [max 5]
9. **Pyruvate** from glycolysis is converted to **acetyl-CoA** via **oxidative decarboxylation**;
10. Krebs cycle involves 3 main stages:
11. Acetyl CoA (2C) joins the cycle by combining with **oxaloacetate** (4C) to form **citrate** (6C);
12. Citrate is **decarboxylated** and **oxidised by dehydrogenation** to form **α-ketoglutarate** (5C) and **NADH;**

This process is also an **oxidative decarboxylation** reaction;

13. **Oxaloacetate** (4C) is regenerated;
14. Krebs cycle involves multiple reactions, including 1 substrate-level phosphorylation, 1 decarboxylation, and 3 dehydrogenation reactions;
15. As a result 1 ATP, 1 CO_2, 2 NADH, and 1 $FADH_2$ are produced per molecule of α-ketoglutarate;
16. For **each** molecule of glucose, glycolysis yields **2 molecules of pyruvate**, and hence **2 molecules of acetyl CoA**. As such, **two rounds of Krebs cycle** are needed to **completely oxidise one molecule of glucose**;

17. All 6 carbon atoms in glucose are lost as 6 CO_2;
18. For each 6C glucose molecule, **2 CO_2** are released via **link reaction**, and **4 CO_2** are released via **Krebs cycle;**
19. The mobile electron carriers (**NADH and $FADH_2$**) with their reducing power will next be transported to the **electron transport chain**, where the bulk of **ATP** is generated;

C. Electron Transport Chain and Cyclic Photophosphorylation [max 5]

20. The photo-excited **electron from P700** is captured by the **PS I primary electron acceptor;**
21. and then passed on to the middle part of the **first electron transport chain (ETC);**
22. As the photo-excited electron travels down the ETC, which consists of **electron carriers** of **progressively lower energy levels**, energy lost is **coupled to the formation of ATP via chemiosmosis;**
23. This way of synthesizing ATP using light energy is called **cyclic photophosphorylation** (This electron eventually fills the electron "hole" left in P700, completing the cycle);
24. **No NADPH** is produced but passing the excited electrons to the second electron transport chain as in the non-cyclic light-dependent reaction, they are transferred to the first electron transport chain;
25. **No O_2** is produced as there is **no photolysis of water;**
26. Hence, **only ATP** is produced by **cyclic light-dependent reaction;**

D. Calvin Cycle [max 5]

Calvin cycle is a pathway that reduces carbon dioxide to produce carbohydrates. It comprises of 3 phases:

27. Carbon fixation - This step involves **carbon dioxide** combining with **RuBP** (ribulose bisphosphate, a 5C sugar);
28. Product is an unstable 6C intermediate that will immediately split to form 2 molecules of **glycerate phosphate** (for every one molecule of CO_2);
29. PGA reduction - **GP is reduced (gains electrons)** to form a 3C compound, **glyceraldehyde-3-phosphate (G3P)**;
30. This reaction requires the reducing power of **NADPH** and energy of **ATP** (products of non-cyclic light-dependent reaction);
31. RuBP regeneration - G3P has to be used to regenerate **RuBP;**
32. **5** molecules of **G3P** (a 3C molecule; total 15C) used to regenerate **3 RuBP** (a 5C molecule; total 15C);
33. **3 ATP** from the light-dependent reaction is used;

E. Polymerase Chain Reaction (PCR) [max 5]

34. A brief heat treatment (up to **95°C**) to **denature** / **unzip and separate** the two strands of DNA double helix;
35. This exposes the bases for complementary base pairing required in subsequent steps;
36. Cooling of the DNA (to ~**60°C**) in presence of a large excess of **DNA primers** allows their specific **annealing** to complementary sequences at the 3' ends of each of the template DNA strand;
37. *Taq* polymerase **synthesises** the complementary DNA strand by catalysing the formation of **phosphodiester bonds** between dNTPs at an optimum **72°C**;

38. **Chain extension** occurs from 3' end of DNA primer which provides free 3' OH group required by *Taq* polymerase;
39. When the above elongation is completed, the primer has been lengthened into a **new complementary strand** of the single-stranded DNA fragment;
40. Because both separated DNA strands are used as templates, there are now **two copies of the original fragment** (each DNA molecule becomes two);
41. Each cycle results in a **doubling** in number of DNA molecules being replicated;
42. The amount of desired sequence hence **increases exponentially** after **multiple cycles**;

QWC: At least 3 accounts of relevant cycles, clearly indicating the cyclical nature, importance or significance of the cycle.

Possible AVPs:

Feedback cycle / Enzyme feedback inhibition

5 **(a)** Explain how anatomical and molecular homology support Darwin's theory of evolution on descent with modification. [10]

DEFINITION

1. **Homology** refers to **similar characteristics** (anatomical or molecular) found in **different species** due to **common ancestry**;

ANATOMICAL HOMOLOGY (max 7m)

2. Organisms with **anatomical homology** have **anatomical (A: morphological) structures** such as bones, organs and gross structural features that they **share with a common ancestor** and thus supports Darwin's theory of descent with modification;
3. Named e.g.: ***pentadactyl limb**** structure in forelimbs;
4. of all **tetrapods/humans, cats, whales** (or any valid e.g. where they state a few organisms or a group of organisms);
5. Forelimbs have **same arrangement of bones** but have **different functions** and superficially look different;
6. e.g. **legs for walking** in cats, **flippers for swimming** in whales (any two e.g.);
7. 5 digit pentadactyl limb structure in common ancestor was altered by **natural selection/different selection pressures** in different organisms to suit **specialised functions/environments**, resulting in variations of the pentadactyl limb structure;
8. Anatomically homologous structures that are **greatly reduced in size** / **have little to no function** as they were **selected against**;
9. are called ***vestigial structures****;
10. Organisms with vestigial structures **share common ancestry** with organisms in which the **structure is still functional** and thus supports Darwin's theory of descent with modification.
11. Named e.g.
    Hind limbs (femur) / hips (pelvic bones) in whales are reduced to **small bones** (i.e. vestigial structures) as they are no longer beneficial to whales which swim. However, their presence in whales **suggest common ancestry with tetrapods (terrestrial ancestors with four legs)**;
    Appendix in humans is also a vestigial structure as it is **reduced** from cecum of its **primate ancestors** which was involved in digestion of plant material. Thus, presence of appendix in humans, **suggests common ancestry with primates**;
    Fossil records: modification of homologous structures with time

MOLECULAR HOMOLOGY

12. Organisms with **molecular homology** have **similar *DNA, RNA & amino acid** sequences** as they **share a common ancestor** that had these molecules;
13. Named e.g. **cytochrome C/ p53 / haemoglobin** are homologous genes;
14. **Homologous genes** share significant **sequence homology** and when expressed, produce **proteins** that have **same function** in all organisms that possess them and thus supports Darwin's theory of descent with modification;
15. **Nucleotide sequences** in ancestral genes were modified due to **accumulation of mutations** that occurred over many generations that were selected for;
16. The **greater the sequence similarity** between homologous genes, the **more closely related** the 2 species are;

QWC 1 anatomical and 1 molecular homology

(b) Explain the significance of different biomolecular composition in membranes of different cells and different organelles. [15]

Biomolecules in membrane:
Phospholipids, proteins, glycolipids, glycoproteins, cholesterol

Significance of different biomolecular composition in membrane:
In different organelles (max **10m**)
1. **Mitochondria** and **chloroplasts**: require **proteins**, such as **electron carriers** and **ATP synthase**, to be arranged in order
2. To facilitate **electron transfer along electron transport chain** and for **chemiosmosis** to be carried out for **ATP synthesis**;
3. Ref. to **oxidative phosphorylation** and **photophosphorylation**
4. **Chloroplasts**: **photosystems / photosynthetic pigments** on thylakoid membrane for **absorption of light**;
5. **Nucleus**: require opening such as **nuclear pores** to be present in the membrane for e.g. transport of RNA out of nucleus (at least one e.g.);
6. **Rough endoplasmic reticulum (RER)**: more **channel proteins** for newly synthesized **proteins** to **enter lumen**;
7. **Smooth endoplasmic reticulum (SER)**: more **Ca^{2+} channel proteins** for release of Ca^{2+} into cytosol;
8. **Lysosome**; more **proton pumps** to maintain acidic internal environments;
9. AVP

In different cells (max **10m**)
10. Different cell types contain different **amount** and **types** of **glycoproteins** and **glycolipids** at **cell surface membrane**;
11. For **cell-cell recognition/communication** and **cell-cell adhesion** to form **tissues**
12. Different cells require **specific receptor proteins** to be present on **cell surface membrane**;
13. To allow **specific ligand** (signalling molecule) to recognize and **bind**;
14. For **e.g.** muscle cell with glucagon receptor;
15. Cells that **produce hydrophilic molecules for extracellular use** contains more **transport/channel/carrier proteins** on the **cell surface membrane**;
16. For e.g. ???
17. **Cell surface membrane** of **immune cells**, for eg. **B cell / macrophage / T helper cell**, contain **specific receptor** eg. **BCR** with specific antigen binding sites;
18. That enable it to bind to specific **antigen**;
19. Cells in organisms living in areas of **higher temperatures** have **cell surface membrane** with **lower percentage of unsaturated fatty acids** (A: higher percentage of saturated fatty acids) **in phospholipids / phospholipid bilayer**; (A: reverse for lower temperatures)
20. For **lower membrane fluidity** and **greater membrane stability**; (A: reverse for lower temperatures)
21. **Cell surface membrane** with **varying amounts** of **cholesterol** that help to **regulate membrane fluidity**
22. AVP

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